TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of a Causal Digital Twin for Real-Time Wafer Dicing Quality Control and Yield Optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (V), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (V), Next-gen Communication ( ), Advanced Robotics & Manufacturing (V), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Equipment	Measurement/Inspection Equipment	Software	Software Solutions
Concept of Proposed Technology	We develop a causal digital twin software for real-time wafer dicing monitoring and control. A causal digital twin is a data-linked simulator of the dicing process that predicts how quality will change when we intervene on controllable settings, rather than merely reproducing correlations in past logs. Conventional SPC or ML can predict defect risk for observed settings, but it cannot reliably answer intervention questions such as "If we change feed rate now, how will kerf drift or chipping change in the next lot?" Our twin estimates actionable intervention effects from production logs by linking controllable dicing settings such as spindle speed, feed rate, coolant flow, and blade condition to quality outcomes such as kerf width drift, chipping, and microcracks through in-process sensor signals such as acoustic emission, vibration, motor current, and vision, while accounting for latent tool state and wafer variability. The software provides early-warning risk, root-cause pathway explanations, and recommended parameter adjustments with uncertainty bounds and safety guardrails for operator-in-the-loop control. The deliverable is deployment-ready software with MES/SCADA integration hooks and audit logs for continuous improvement.			

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Rationale for Joint R&D Project	Wafer dicing is a yield-critical step where small drifts in kerf, chipping, or microcracks can create large downstream loss. With 2.5D/3D packaging increasing dicing complexity, legacy SPC cannot keep up with the multivariate interactions. Korean semiconductor SMEs already collect lot-level setpoints, multi-sensor streams, and inspection labels, but SPC and black-box monitors mainly learn associations that can mislead control decisions when tool condition or wafer variability shifts. For example, when vibration rises due to blade wear, operators often reduce feed rate. The logs then show low feed rate co-occurring with high defects, and a correlation-driven model may recommend increasing feed rate during excursions, which is unsafe. In this project, a causal digital twin is a data-driven simulator that separates true drivers from operator responses and then compares candidate setpoint changes and maintenance actions through what-if simulation, with uncertainty bounds for operator planning. This fills a critical gap for Korean equipment makers: moving from 'passive monitoring' hardware to 'active decision' software solutions. Phase 1 delivers retrospective excursion diagnosis and a dashboard. Phase 2 adds streaming integration and operator-in-the-loop validation with KPI tracking.
Keywords	1. Wafer Dicing Process 2. Causal Digital Twin 3. Root-Cause Analysis 4. Real-Time Quality Control 5. Yield Optimization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	This project advances wafer dicing quality control from detecting risk to planning safe corrective actions and supporting decision-making. The goal is deployable software that performs real-time monitoring, explains why an excursion happened, and recommends the next best action, for example adjusting feed rate or coolant flow versus scheduling blade maintenance. The recommendations must come with uncertainty so operators can judge whether an action is safe enough to execute. Our group has already built an end-to-end causal inference pipeline that starts from observational data, infers the most plausible causal structure supported by the data, and then estimates causal effects in a sample-efficient way. This is the foundation behind our ICML work [1], where causal effects can be estimated even when the causal structure is only partially known. In this project, we will turn this capability into a causal foundation model for wafer dicing. The model will ingest streaming dicing logs and continuously update a causal graph, which encodes the model's understanding of physical world (wafer dicing process) and how it is affected by controllable settings and sensor signals, and a set of intervention effect estimates that answer what-if questions in real-time. This foundation model will act as a “decision-grade process map” that remains valid under drift and enables rapid root-cause localization and safe action screening during excursions. For root-cause explanations, we have developed a method for interpreting black box system [2] by quantifying how much each input control causally contributes to a quality outcome. This matters in wafer dicing because correlations in historical logs can reflect operator reactions and latent tool conditions rather than true causal drivers. When vibration rises due to unmeasured blade wear, operators often reduce feed rate. The logs then show low feed rate co-occurring with high defects, even though low feed rate was a response to the underlying problem and may actually reduce defects if blade condition were held fixed. A correlation-based tool can therefore recommend increasing feed rate during excursions, which is unsafe on worn blades. We will extend causal contribution to wafer dicing for root-cause analysis to identify a most significant causal driver of defects. Coupled with the causal foundation model, the system will then simulate candidate interventions on wafer dicing settings and maintenance actions, rank feasible options, and recommend the safest high-impact corrective action under uncertainty. Finally, we will build a causal simulator that remains reliable under hidden confounding, which is prevalent in wafer dicing because latent tool state, wafer batch variability, and unrecorded environment changes affect both settings and outcomes. We will combine two existing foundations from our lab. The first is scalable sensor-mediated causal effect estimation [3] that stays stable with high-dimensional sensor signals. The second is uncertainty-aware causal modeling that produces certified bounds [4] when latent factors prevent precise point estimates. Together, they enable an interventional generative model for wafer dicing that simulates quality outcomes under candidate setpoint changes and maintenance actions, with uncertainty bounds for operator planning. In short, the causal foundation model, causal contribution root-cause analysis, and uncertainty-aware intervention simulator together form the causal digital twin engine that supports real-time wafer dicing QC and beyond QC decision-making.			

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R&D Target Specifications	**Data readiness and integration**: We will build an end-to-end pipeline that aligns, per lot, (i) recipe and setpoints, (ii) sensor measurements available at the SME (e.g., vibration and motor current, or acoustic and vision), and (iii) downstream inspection outcomes (kerf drift and at least one defect label such as chipping or microcracks). The target is successful alignment for at least 80% of lots in the chosen pilot scope, with missingness patterns and quality issues explicitly documented. **Operational latency**: In shadow-mode operation, the system will output monitoring signals and root-cause explanations at lot cadence, meaning updated results are available within one lot cycle. For interactive what-if queries used during troubleshooting, the target is response within a few seconds for typical cases on the pilot dataset and compute environment. **Monitoring usefulness**: Relative to the SME baseline (current SPC rules and/or existing black-box monitor), the system will reduce nuisance alarms at a matched detection level on a time-based holdout period. If the baseline does not have a clear detection metric, we will hold the alarm volume fixed and increase the fraction of alarms that correspond to inspection-confirmed quality degradation. We will report performance with time-split evaluation and uncertainty (e.g., confidence intervals) rather than relying on single-point improvements. **Root-cause explanation quality**: For excursions where a plausible cause label is available from maintenance records, tool-change events, or engineer annotations, the system will rank the validated driver within its top-3 suspected drivers in at least 60% of held-out cases. Every explanation will include an evidence trail (settings and sensor-mechanism pathway) and will be stored for engineering review. **Action recommendations and feasibility**: For each excursion, the system will output a short list of recommended actions that includes both setpoint adjustments and maintenance actions when applicable. The top-ranked recommendation will be actionable and compliant with SME-provided operating constraints (recipe ranges and safety rules) in at least 70% of validation cases. Any recommendation that violates a hard safety constraint will be automatically blocked. **Uncertainty and trust**: For key outcomes used in decision-making (e.g., kerf drift and chipping risk), the system will provide uncertainty bounds. The bounds will be calibrated so that, on held-out time periods, the realized outcomes fall within the reported bounds at least 90% of the time, while keeping the bounds practically informative rather than trivially wide. **Pilot impact**: In Phase 2, we will target at least one modest, verifiable improvement in the pilot scope, such as a 5–10% reduction in scrap or rework for a targeted defect mode, or a 20% reduction in mean time-to-diagnosis for excursions, or an agreed measurable yield improvement on pilot lots. The KPI will be selected jointly based on what the SME already tracks reliably and can validate during a staged rollout. **Auditability**: 100% of monitoring outputs, root-cause explanations, and recommendations will be logged with timestamps, input summaries, and version identifiers to support engineering review, traceability, and continuous improvement.
R&D Plan	We structure the plan around three technical pillars that together implement the causal digital twin engine: (i) causal foundation model, (ii) causal-contribution root-cause analysis, and (iii) uncertainty-aware intervention simulator for action planning. In Phase 1 (Months 0–6), we will leverage retrospective data from historical lots. **Task 1. Data onboarding and alignment.** Build the unified schema for lot-level setpoints, sensor streams, and inspection outcomes; define defect taxonomy and excursion definitions with the SME. Deliver a reproducible ETL pipeline and data quality report.

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	Task 2. Baselines and evaluation protocol. Implement baseline SPC and any existing predictors; define fixed evaluation splits (time-based) and KPI definitions, including "validated cause" rules using maintenance logs. **Task 3. Prototype causal foundation model (offline).** Train an initial causal graph and intervention-effect estimator that separates driver variables (tool condition proxies, settings) from response variables (operator reactions). Produce "what-if" estimates for a short list of controllable knobs and maintenance actions. **Task 4. Prototype root-cause explanations.** Implement causal-contribution explanations for the top defect modes; validate Top-k root-cause accuracy on labeled excursions. At the end of Phase 1, we aim that a working dashboard that supports retrospective excursion diagnosis, root-cause ranking, and offline what-if queries on historical lots. In Phase 2 (Months 7–36), we will leverage real-time data, aiming to deploy the causal digital twin model. **Task 5. Streaming causal foundation model.** Convert the offline model into an online-updating model that refreshes causal structure and effect estimates under drift (tool wear, wafer mix). Add drift monitors and automatic recalibration triggers. **Task 6. Uncertainty-aware intervention simulator.** Build an interventional generative simulator that predicts outcomes under candidate setpoint changes and maintenance actions, returning uncertainty bounds and rejecting unsafe actions using hard constraints. **Task 7. Action recommendation layer.** Implement action ranking that balances expected quality improvement, uncertainty, and operational cost. Provide operator-in-the-loop UI flows and audit logging. **Task 8. Deployment and staged pilot.** Integrate with MES and SCADA in "shadow mode" first, then proceed to limited operator-guided trials. Track KPIs continuously and run post-mortems to refine the model. At the end of Phase 2, we aim that a deployed causal digital twin model that supports real-time streaming, uncertainty-aware intervention simulation, and operator-in-the-loop action recommendation.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Causal Data Science Lab at UIUC will lead the core algorithmic research and software development for the causal digital twin engine, leveraging the lab's expertise in pragmatic causal inference (addressing unmeasured confounding and complex data-generating processes), scalable causal methods for high-dimensional sensor streams, and causal decision-making for safe action recommendations under uncertainty. The lab will design and implement the causal modeling framework that estimates intervention effects from observational production logs while distinguishing true process drivers from operator responses and latent variables. Deliverables include prototype software modules covering the inference engine, explanation layer (causal-contribution root-cause analysis), and dashboard with API interfaces. The lab will provide technical documentation, validation reports benchmarking performance against SME baselines, and training sessions to ensure partner engineers can interpret uncertainty bounds and recommendation logic.
Expected R&R of Korean SME Partner	The Korean SME partner will provide access to production-grade wafer dicing tools and the associated production data infrastructure, including synchronized datasets of recipe and setpoint logs, multi-sensor measurements (for example, vibration, motor current, acoustic emission, or vision), and downstream inspection outcomes covering kerf drift and defect labels such as chipping and microcracks. The SME will lead defect taxonomy definition, operational constraint specification (recipe ranges, safety rules, tooling limits), and pilot testbed governance, including approvals for phased rollout from shadow mode to operator-in-the-loop trials. The SME will implement MES and SCADA integration, ensuring that UIUC-developed modules can receive real-time data streams and push recommendations back into existing workflow systems. The SME will lead operator-in-the-loop validation by coordinating with process engineers and shift supervisors to test recommendations in controlled settings, documenting feasibility and outcomes. Beyond the technical pilot, the SME will take ownership of deployment engineering, including hardening the software for factory environments, scaling to additional tools or product lines, and executing the commercialization strategy for broader adoption within the SME and potentially to other semiconductor manufacturers.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical outcomes:** Phase 1 delivers a retrospective causal digital twin dashboard that supports root-cause diagnosis, what-if querying, and action screening on historical excursions, with validated top-3 root-cause recall of at least 60% and actionable recommendations in at least 70% of cases. Phase 2 delivers a deployed, streaming causal digital twin that performs real-time monitoring, provides uncertainty-calibrated root-cause explanations and intervention recommendations with 90% coverage bounds, and supports operator-in-the-loop trials with full audit logging. **Manufacturing impact:** The system will reduce nuisance alarms while maintaining or improving detection coverage, provide faster root-cause diagnosis with evidence trails, and enable safer corrective actions backed by uncertainty bounds. In Phase 2 pilot trials, the target is at least one modest, verifiable improvement such as a 5-10% reduction in scrap or rework, a 20% reduction in mean time-to-diagnosis, or a measurable yield gain on pilot lots, selected jointly with the SME based on what can be reliably tracked. **Commercial and deployment value:** The deliverable is a deployment-ready software module with MES and SCADA integration hooks, designed for Korean semiconductor SME factories and extensible to related dicing tools, wafer-level process steps, and other precision manufacturing applications where causal decision support under uncertainty is critical.

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	Broader economic value: Higher process reliability, reduced scrap and rework costs, and faster excursion recovery for Korean SME semiconductor supply-chain players, supporting competitiveness in yield-critical backend processes.

6. References

[1] **Jung, Y.**, Tian, J., and Bareinboim, E. "Estimating identifiable causal effects on Markov equivalence class through double machine learning." International Conference on Machine Learning (ICML), 2021.

[2] **Jung, Y**., Kasiviswanathan, S., Tian, J., Janzing, D., Blöbaum, P., and Bareinboim, E. "On measuring causal contributions via do-interventions." International Conference on Machine Learning (ICML), pp. 10476–10501, 2022.

[3] **Jung, Y.** "Debiased front-door learners for heterogeneous effects." International Conference on Learning Representations (ICLR), 2026.

[4] **Jung, Y.** and Kang, B. "Information-Theoretic Causal Bounds under Unmeasured Confounding." arXiv preprint arXiv:2601.17160v2, 2026.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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